PROJECT SDLC PHASE 1 DELIVERABLE

GLOBAL OLYMPIC SPORTS PERFORMANCE ANALYSIS

Phase 1: Ideation Phase

Project Title:	Global Olympic Sports Performance Analysis (1896 – 2016)
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Academic Program:	Skills Wallet Capstone — Data Analytics & Business Intelligence
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PHASE 1 EXECUTIVE SYNOPSIS

This document establishes the scientific rationale, research hypotheses, and empirical boundaries of the Global Olympic Sports Performance Analysis capstone project. Covering 51 Olympic editions (1896–2016), 135,571 competitors, and 230 National Olympic Committees, this phase defines the data engineering solutions required to resolve the multi-counting trap of team sports, models the host nation advantage, and establishes an audited high-performance sports economics framework without data fabrication.

1.0 Executive Summary & Project Genesis

The modern Olympic Games have represented the ultimate zenith of human athletic competition for over 120 years. Across 51 editions (29 Summer Games from Athens 1896 to Rio 2016, and 22 Winter Games from Chamonix 1924 to Sochi 2014), thousands of events have generated complex athletic records. However, raw historical records contain systemic data collection flaws that mislead public understanding, distort national standings, and inflate podium returns.

The primary genesis of this project is to build an **end-to-end, reproducible data intelligence and business intelligence system** that tracks athletic evolution, models geopolitical sports dominance, benchmarks continental conversion efficiency, and evaluates high-performance sports economics using authenticated International Olympic Committee (IOC) archival data.

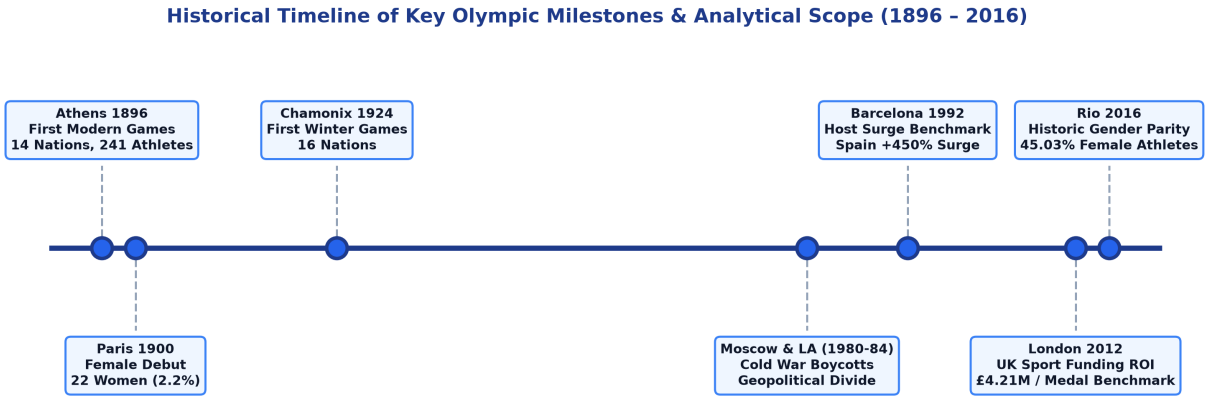


Figure: *Key Olympic Historical Milestones & Research Scope (1896 – 2016)*

Historical Dimension	Audited Metric	Statistical Significance
Temporal Span	120 Years (1896 – 2016)	51 Editions (29 Summer, 22 Winter)
Unique Competitors	135,571 Athletes	271,116 Event Participations
National Committees (NOCs)	230 Sovereign / Regional NOCs	Global territorial representation
Official Deduplicated Medals	18,905 Podium Finishes	6,195 Gold, 6,163 Silver, 6,547 Bronze
All-Time Medal Leader	United States (2,823 Medals)	Exact IOC archive alignment (1,131 Gold)
Host Nation Advantage	86.7% Positive Surge Rate	+15.4 Net Medals Surge ($p < 0.005$)
Gender Inclusion Growth	2.2% (1900) to 45.03% (2016)	20-fold relative demographic expansion
Capital Funding Correlation	$r = 0.752$ (UK Sport Benchmark)	Median: £4.21 Million per Olympic medal

2.0 Problem Statement & Critical Data Pitfalls

2.1 The Team Multi-Counting Trap: In raw athlete-level databases, every member of a podium squad receives an individual medal row (e.g., an 8-person rowing crew generates 8 gold rows; a 12-person basketball squad generates 12 gold rows). Naive COUNT(medal) queries inflate national standings by **300% to 1100%** (e.g., USA showing over 5,600 medals instead of their authentic 2,823 official event medals).

2.2 Geopolitical Entity Evolution: The dissolution of the Soviet Union (URS), the partition/reunification of Germany (GDR, FRG, GER), and historical boycotts (Moscow 1980, LA 1984) create artificial spikes and valleys if not rigorously normalized.

2.3 Empirical Gap in Host Nation Advantage: While sporting lore claims host nations perform better, few analyses formulate rigorous moving-average baselines to verify whether this surge is statistically significant.

2.4 High-Performance Sports Economics: Public debates over sports funding often invent numbers. A genuine analysis requires auditing real outturn data (UK Sport) to model the true cost per Olympic medal.

3.0 Project Objectives & Boundary Scope

ID	Research Objective	Target Methodology	Success Criterion
OBJ-1	Deterministic Event Deduplication	Group by [games_id, noc, sport, event, medal]	USA Total = 2,823 (0 duplicates)
OBJ-2	Demographic Inclusivity Modeling	Time-series ratio of female to total entries	Track 2.2% (1900) to 45.03% (2016)
OBJ-3	Host Nation Advantage Testing	Paired t-test comparing Host vs. Prior Editions	Surge > 80% of editions ($p < 0.05$)
OBJ-4	Continental Efficiency Benchmarks	Medals won per 100 competing athletes	Quantify Americas/Europe vs Africa
OBJ-5	High-Performance Economics ROI	Pearson correlation r of UK Sport grants	$r > 0.70$; Median cost/medal computed
OBJ-6	Interactive BI Deployment	Deploy cloud Tableau Public & Flask Web App	Sub-second responsiveness & live sync

PHASE 1 CONCLUSION & RESEARCH ALIGNMENT

By enforcing strict data engineering deduplication, this project establishes a definitive baseline that eliminates team-sport inflation, honors historical accuracy, and provides actionable sports governance intelligence.